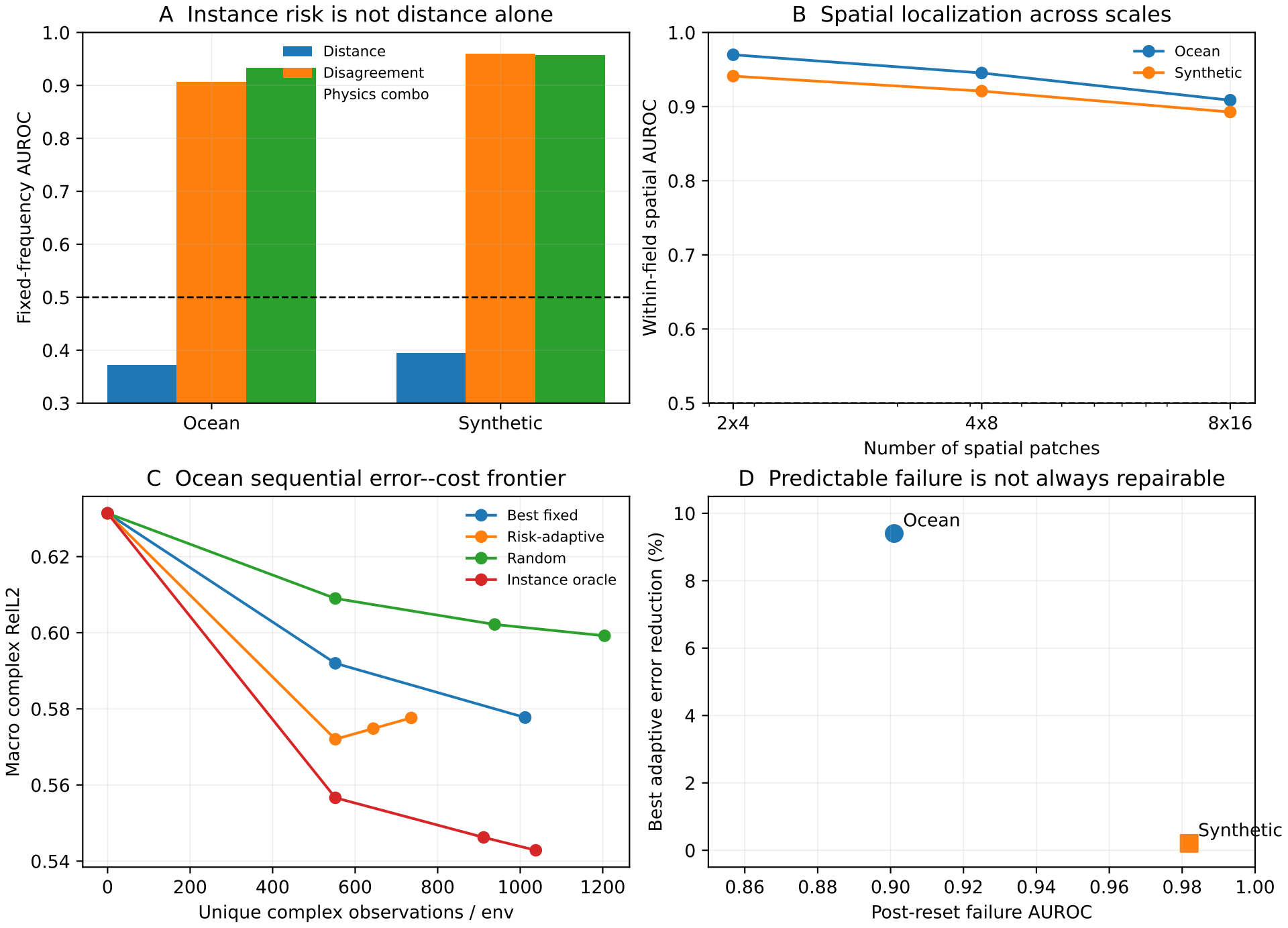
A Instance risk is not distance alone
Fixed-frequency AUROC
Distance
Disagreement
Physics combo
Ocean
Synthetic
B Spatial localization across scales
Within-field spatial AUROC
Ocean
Synthetic
2x4
4x8
8x16
Number of spatial patches
C Ocean sequential error--cost frontier
Macro complex RelL2
Best fixed
Risk-adaptive
Random
Instance oracle
Unique complex observations / env
D Predictable failure is not always repairable
Best adaptive error reduction (%)
Ocean
Synthetic
Post-reset failure AUROC